Domain Name System – Performance tests
29 April 2026

Compare the performance of different Name Servers varying the transport protocols (i.e., UDP, TCP, TLS, HTTPS, HTTP/3, QUIC) and the query types (e.g., A, NS, ANY, MX) using measurement tools such as **`dnsping`**, **`dnseval`** and **`dnsperf`**.

```
[mcc@ercole ~]$ dnsping
dnsping version 2.9.3
Usage: dnsping [-346aDeEFhLmqnrvTQxXH] [-i interval] [-w wait] [-p dst_port] [-P src_port] [-S src_ip]
               [-c count] [-t qtype] [-C class] [-s server] [--ecs client_subnet] hostname

  -h, --help        Show this help message
  -q, --quiet       Suppress output
  -v, --verbose     Print the full DNS response
  -s, --server      Specify the DNS server to use (default: first entry from /etc/resolv.conf)
  -p, --port        Specify the DNS server port number (default: 53 for TCP/UDP, 853 for TLS)
  -T, --tcp         Use TCP as the transport protocol
  -X, --tls         Use TLS as the transport protocol
  -H, --doh         Use HTTPS as the transport protocol (DoH)
  -3, --http3       Use HTTP/3 as the transport protocol (DoH3)
  -Q, --quic        Use QUIC as the transport protocol (DoQ)
  -4, --ipv4        Use IPv4 as the network protocol
  -6, --ipv6        Use IPv6 as the network protocol
  -P, --srcport     Specify the source port number for the query (default: 0)
  -S, --srcip       Specify the source IP address for the query (default: default interface address)
  -c, --count       Number of requests to send (default: 10, 0 for unlimited)
  -r, --norecurse   Enforce a non-recursive query by clearing the RD (recursion desired) bit
  -m, --cache-miss  Force cache miss measurement by prepending a random hostname
  -w, --wait        Maximum wait time for a reply (default: 2 seconds)
  -i, --interval    Time interval between requests (default: 1 second)
  -t, --type        DNS request record type (default: A)
  -L, --ttl         Display the response TTL (if present)
  -C, --class       DNS request record class (default: IN)
  -a, --answer      Display the first matching answer in rdata, if applicable
  -e, --edns        Enable EDNS0 and set its options
  -E, --ede         (Ignored - EDE messages now always displayed when present)
  -n, --nsid        Enable the NSID bit to retrieve resolver identification (implies EDNS)
      --cookie      Display EDNS cookies when present (implies EDNS)
  -D, --dnssec      Enable the DNSSEC desired flag (implies EDNS)
      --ecs         Set EDNS Client Subnet option (format: IP/prefix, e.g., 192.168.1.0/24) (implies EDNS)
  -F, --flags       Display response flags
  -x, --expert      Display additional information (implies --ttl, --flags)
```

Examples:

`dnsping -c 3 -m -t AAAA www.google.com`

`dnsping -s 9.9.9.9 -v -c 3 -Q -t NS abc.io`

```
[mcc@ercole ~]$ dnseval
dnseval version 2.9.3
Usage: dnseval [-ehmvCTXHQ3SD] [-f server-list] [-j output.json] [-c count] [-t type] [-p port] [-w wait]
hostname

  -h, --help          Display this help message
  -f, --file          Specify a DNS server list file to use (default: system resolvers)
  -c, --count         Number of requests to send (default: 10)
  -m, --cache-miss    Force a cache miss measurement by prepending a random hostname
  -w, --wait          Set the maximum wait time for a reply in seconds (default: 2)
  -t, --type          Set the DNS request record type (default: A)
  -T, --tcp           Use TCP as the transport protocol instead of UDP
  -X, --tls           Use TLS as the transport protocol (DoT)
  -Q, --quic          Use QUIC as the transport protocol (DoQ)
  -H, --doh           Use HTTPS as the transport protocol (DoH)
  -3, --http3         Use HTTP/3 as the transport protocol (DoH3)
  -j, --json          Save the results to a specified file in JSONL format (one JSON object per line)
  -p, --port          Specify the DNS server port number (default: protocol-specific)
  -S, --srcip         Set the query source IP address
  -e, --edns          Enable EDNS0 in requests
  -D, --dnssec        Enable the 'DNSSEC desired' (DO flag) in requests
  -C, --color         Enable colorful output
  -v, --verbose       Print the full DNS response details
      --skip-warmup   Disable cache warmup (default: warmup enabled)
```

Examples:

`dnseval -f Name-Servers -v -c 3 -m -t AAAA www.google.com`

`dnseval -f Name-Servers -v -c 3 -Q -t NS abc.io`

It is necessary to create a file with the list of the Name Servers to be tested.

```
[mcc@ercole ~]$ dnsperf -h
Usage: dnsperf [-f family] [-m mode] [-s server_addr] [-p port]
               [-a local_addr] [-x local_port] [-d datafile] [-c clients]
               [-T threads] [-n maxruns] [-l timelimit] [-b buffer_size]
               [-t timeout] [-e] [-E code:value] [-D]
               [-y [alg:]name:secret] [-q num_queries] [-Q max_qps]
               [-S stats_interval] [-u] [-B] [-v] [-W] [-h] [-H] [-O]
  -f address family of DNS transport, inet or inet6 (default: any)
  -m set transport mode: udp, tcp, dot or doh (default: udp)
  -s the server to query (default: 127.0.0.1)
  -p the port on which to query the server (default: udp/tcp 53, DoT 853 or DoH 443)
  -a the local address from which to send queries
  -x the local port from which to send queries (default: 0)
  -d the input data file (default: stdin)
  -c the number of clients to act as
  -T the number of threads to run
  -n run through input at most N times
  -l run for at most this many seconds
  -b socket send/receive buffer size in kilobytes
  -t the timeout for query completion in seconds (default: 5)
  -e enable EDNS 0
  -E send EDNS option
  -D set the DNSSEC OK bit (implies EDNS)
  -y the TSIG algorithm, name and secret (base64)
  -q the maximum number of queries outstanding (default: 100)
  -Q limit the number of queries per second
  -S print qps statistics every N seconds
  -u send dynamic updates instead of queries
  -B read input file as TCP-stream binary format
  -v verbose: report each query and additional information to stdout
  -W log warnings and errors to stdout instead of stderr
  -h print this help
  -H print long options help
  -O set long options: <name>=<value>
```

A man page with the details of the command is also available (i.e., man dnsperf).

Examples:

`dnsperf -d queries -s 193.204.35.27 -v`

`dnsperf -d queries -s 1.1.1.1 -v -m dot`

It is necessary to create a file which includes the queries to the sent to the Name Server specified in the command line.